Kordel's

Glucosamine Chondroitin MSM

Effervescent Tablet

COMPOSITION
Each tablet contains Glucosamine Sulfate 750 mg (as Glucosamine Sulfate Potassium Chloride Complex 995mg), Chondroitin Sulfate 600 mg (as Chondroitin Sulfate Sodium 666mg) and Methylsulfonylmethane 300 mg.

INDICATION
As an adjuvant therapy for osteoarthritis.

PHARMACOLOGY AND MECHANISM OF ACTION
Glucosamine is a natural substance found in chitin, mucoproteins, and mucopolysaccharides. It is involved in the manufacture of glycosaminoglycan, which forms cartilage tissue in the body; glucosamine is also present in tendons and ligaments. Glucosamine must be synthesised by the body but the ability to do this declines with age. Glucosamine and its salts have therefore been advocated in the treatment of rheumatic disorders including osteoarthritis.

Glucosamine also acts to improve the viscosity of synovial fluid by increasing synovial fluid production, thereby providing lubricant activity.

Chondrotin sulfate, as a vital component of aggrecan, helps to maintain the structural integrity of cartilage tissue. Loss of chondroitin sulphate from joint cartilage leads to loss of cartilage resilience and shock-absorbing properties. Orally administered chondroitin sulphate leads to increases in hyaluronic acid production and viscosity of synovial fluid. Chondroitin sulphate also inhibits enzymes involved in degradation of cartilage.

Methylsulfonylmethane, or MSM is a naturally occurring organic sulphur-containing compound. It has been shown to have anti-inflammatory properties, especially when used in combination with glucosamine and chondroitin.

MSM helps to support the natural formation of cartilage and connective tissues needed for healthy joint function and mobility. MSM is a source of bio-available sulphur that provide additional helps to nourish the lubricating fluid in the joints.

Combination therapy with glucosamine, chondroitin and MSM helps to protect joint cartilage from damage (chondroprotection), reduces joint pain and improves joint function and mobility.

PHARMACOKINETICS
Absorption
After oral administration, bioavailability is low due to first-pass hepatic metabolism~26%.
The gastrointestinal absorption is close to 90%.

Distribution
Glucosamine is not protein-bound, but rather incorporates into plasma proteins (primarily globulins).
Volume of Distribution: 2.5 Liters.

Metabolism
- Liver, extensive.
The first-pass effect in the liver in which more than 70% of glucosamine is metabolized.

Excretion
Renal Excretion, 10%.
Feces, 11%.
Part of a dose of glucosamine sulfate is eliminated as carbon dioxide via expired air.

As for Chondroitin, pharmacokinetic studies performed on humans and experimental animals after oral administration of chondroitin sulfate revealed that it can be absorbed orally. Chondroitin sulfate shows first-order kinetics up to single doses of 3,000 mg. Multiple doses of 800 mg in patients with osteoarthritis do not alter the kinetics of chondroitin sulfate. The bioavailability of chondroitin sulfate ranges from 15% to 24% of the orally administered dose. More particularly, on the articular tissue, Ronca et al reported that chondroitin sulfate is rapidly absorbed in the gastro-intestinal tract and a high content of labeled chondroitin sulfate is found in the synovial fluid and cartilage.

As with Glucosamine, the pharmacokinetic profile and distribution of MSM is derived primarily from animal studies. Following a single oral dose of 500mg/kg, MSM was rapidly and efficiently absorbed with a mean t max of 2.1 h, Cmax of 622 μg equiv/mL, and AUC0-inf of 15124 h.μg equiv/mL. The t1/2 was 12.2 h. Soft tissue distribution indicated a fairly homogeneous distribution throughout the body with relatively lower concentrations in skin and bone. Approximately 85.8% of the dose was recovered in the urine after 120 h, whereas only 3% was found in the feces. No quantifiable levels of MSM were found in any tissues after 120 h, indicating complete elimination of MSM. The results of the study suggest that MSM is rapidly absorbed, well distributed, and completely excreted from the body.

DOSAGE
Tablet should be taken 15-30 minutes before meal.
<u>Mild or moderate osteoarthritic symptoms:</u>
1-2 tablets daily for at least 8 weeks.

<u>Severe osteoarthritic symptoms:</u>
Initial therapy: 1 tablet 2 times daily for at least 3 weeks.
Follow-up therapy: 1 tablet daily for additional 3-4 months.

Kordel's Glucosamine + Chondroitin + MSM Effervescent Tablet is used as an adjunct therapy in the treatment of osteoarthritis. The treatment of osteoarthritis should be repeated every 6 months or less as directed by medical professional.

Pain relief can be seen only after about 1-2 weeks. Therefore, it may be necessary to take an analgesic or anti-inflammatory medication to relief pain during the first few days of therapy.

<u>Children</u>
Safety and effectiveness have not been established in children.

MODE OF ADMINISTRATION
Oral

CONTRAINDICATION
Hypersensitivity to glucosamine or to any of the excipients
As the active ingredient is obtained from seafood (shellfish), the product should not be given to patients who are allergic to shellfish.

WARNING AND PRECAUTIONS
- Glucosamine treats the underlying cause of osteoarthritis and the therapeutic effect can only be seen after 2-3 weeks. Therefore, it is advisable to take an analgesic / anti-inflammatory drug if required during the first 2-3 weeks of therapy with glucosamine.
- Administration during the first three months of pregnancy must be avoided.
- Safety and effectiveness have not been established in children therefore children, should avoid using glucosamine.
- The administration in patients with severe hepatic or renal insufficiency should be made under medical supervision.
- Derived from Seafood, therefore should not be given to patients who are allergic to shellfish.
- A doctor should be consulted in order to exclude the presence of other joint conditions / diseases for which an alternative treatment should be considered.

Effects on Ability to Drive and Use Machines.
No effects on the ability to drive or to operate machines are expected.

INTERACTIONS WITH OTHER MEDICAMENTS
Effects on glucose metabolism & antidiabetic agents:
It has been hypothesized that glucosamine may impair insulin secretion through competitive inhibition of glucokinase in pancreatic beta cells and / or alteration of peripheral glucose uptake.

Glucosamine may increase insulin resistance and consequently affect glucose tolerance.

It may reduce antidiabetic agent effectiveness eg when used with these antidiabetic agent :
Acarbose, Acetohexamide, Chlorpropamide, Glipizidede, Glyburide, Metformin, Miglitol, Pioglitazone, Repaglinide, Rosiglitazone, Glimepiride , Tolbutamide, Troglitazone,
Glucosamine is likely safe in patients with well-controlled diabetes (HbA1c less than 6.5%) taking one or two oral antidiabetic medications or controlled by diet only. In patients with higher HbA1c levels or those taking insulin, monitor blood glucose levels closely/more frequently
Reduced effectiveness when used with glucosamine:
Doxorubicin, Etoposide, Teniposide.

Warfarin
- Elevations of International Normalized Ratio serum values and potentiation of anticoagulant effects.
- If concomitant therapy is necessary, the patient's INR should be more closely monitored.

PREGNANCY AND LACTATION
Available evidence is inconclusive or inadequate for use in pregnant or lactating mothers. Until more information is available, this product should only be used under medical supervision in pregnancy and lactating mothers if the potential benefit to the mother justifies the potential risk to the fetus. Administration during the first 3 months of pregnancy must be avoided.

SIDE EFFECTS
Some individuals may experience gastrointestinal discomforts such as constipation, diarrhoea, dyspepsia, nausea and vomiting which are often mild in nature and disappear on continuation of therapy.

Cardiovascular: Peripheral oedema and tachycardia have been reported in a few patients following larger clinical trials investigating oral administration of glucosamine in osteoarthritis. Causal relationship has not been established.

Central nervous system: Drowsiness, headache, insomnia have been observed rarely in individuals taking glucosamine products (less than 1%).

Gastrointestinal: Nausea, vomiting, diarrhoea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described during oral therapy with glucosamine.

Skin: Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

SYMPTOMS AND TREATMENT OF OVERDOSE
Not known. In case of overdosage, monitor his vital signs and treat symptomatically.

STORAGE CONDITION
Store in a dry place below 30°C.

Keep out of reach of children.
Jauhkan daripada kanak-kanak.
Unsuitable for phenylketonurics.

Shelf life: Please see label.
Packaging available:
10 tablets in round HDPE tube with plastic snap-on cap, packed in outer carton - 10, 20, 30, 40, 50, 60.
15 tablets in round HDPE tube with plastic snap-on cap, packed in outer carton – 15, 30, 45, 60.
Not all pack sizes are marketed.

PRODUCT REGISTRATION HOLDER:
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